

FULL NAME OF STUDENT

BIOLOGY

Bridge Program

Quarter 2

Summative Assessment

**** Practice Questions ****

1 hour

Instructions to student

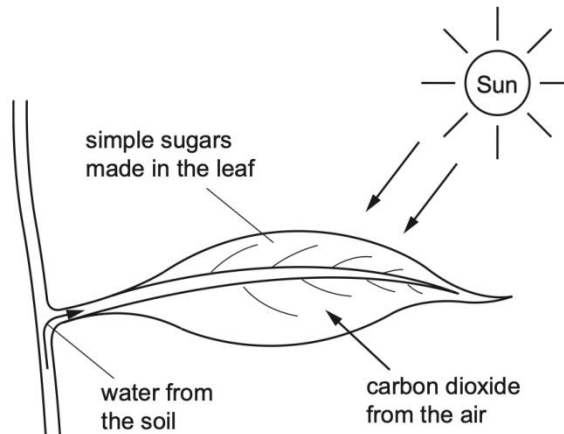
- Write your **full name (passport name)** in the box above using **CAPITAL** letters.
- Do not open this examination paper until instructed to do so.
- The examination paper consists of **two** sections:
 - **Section A** is multiple choice questions;
 - **Section B** is short answer questions.
- Read the instructions for each section carefully.
- Answer **all** questions.
- A calculator is permitted for this paper.
- The maximum mark for this examination paper is [**44 marks**].

10 pages

Section A Multiple Choice Questions

- Answer **all** questions. Each question is [1 mark].
- Choose **one** answer you consider to be the best.
- Indicate your choice in the **answer sheet** (separate from the exam booklet).

- 1 The diagram shows a leaf on a plant.



Which characteristic of life is represented by this diagram?

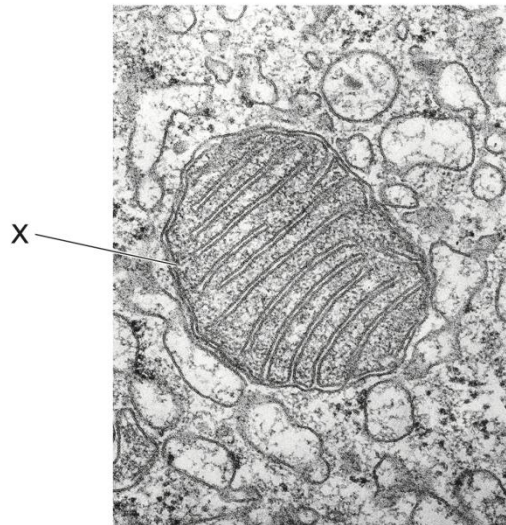
- A** excretion **B** nutrition **C** photosynthesis **D** respiration
- 2 The table shows the scientific names and the common names of four plants.

scientific name	common name
<i>Althaea officinalis</i>	marsh mallow
<i>Hottonia palustris</i>	water violet
<i>Viola palustris</i>	marsh violet
<i>Viola tricolor</i>	wild pansy

Which word is a genus name?

- A** marsh
B *palustris*
C *Viola*
D wild
- 3 A cell is 65 μm across. What is this measurement in mm?
- A** 0.00065 mm **B** 0.0065 mm **C** 0.065 mm **D** 0.65 mm

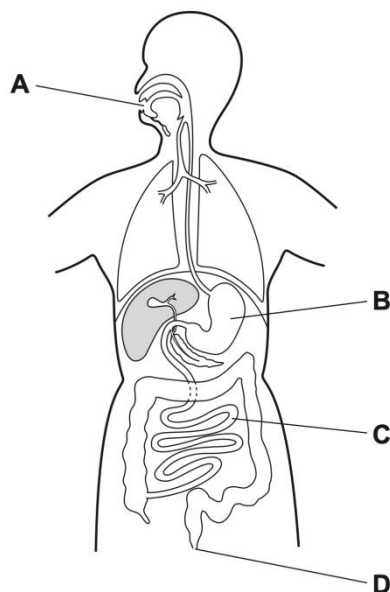
- 4 The photomicrograph shows part of an animal cell.



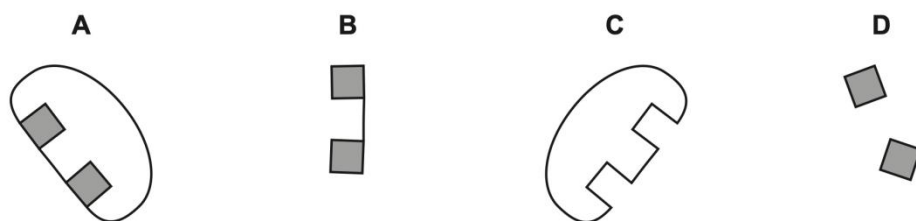
Which biological process occurs within this cell part?

- A** gas exchange **B** photosynthesis **C** protein synthesis **D** respiration
- 5 What increases the rate of diffusion of oxygen into a mammalian cell?
- A** decreasing the concentration gradient
B decreasing the surface area
C increasing the cell membrane thickness
D increasing the temperature
- 6 What is lacking in the diet of a person with scurvy?
- A** calcium **B** iron **C** vitamin C **D** vitamin D
- 7 Large food molecules are built from small molecules called monomers. Which statement is correct?
- A** Amino acids are the monomers of carbohydrates.
B Fibrin is the monomers of fibre.
C Glycerol is a monomer of oils.
D Simple sugar is the monomer of protein.
- 8 Which type of biological molecule are enzymes made of?
- A** carbohydrate
B fat
C oil
D protein

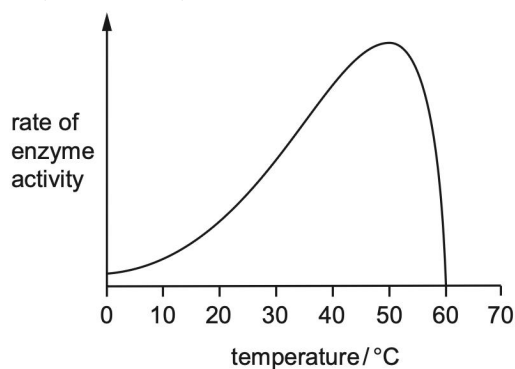
- 9 The diagram shows the digestive system. Where does egestion take place?



- 10 The diagrams show molecules involved in the action of a digestive enzyme. What is the substrate?



- 11 The graph shows how the activity of an enzyme varies with temperature.



What is the optimum temperature for this enzyme and at which temperature is the enzyme completely denatured?

- A 30 °C and 0 °C
- B 30 °C and 60 °C
- C 50 °C and 0 °C
- D 50 °C and 60 °C

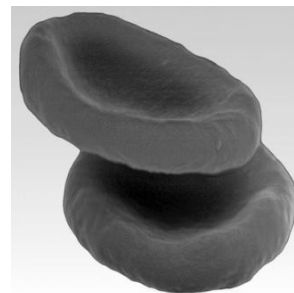
- 12 Enzyme **R** digests protein in the human stomach. Four test tubes are set up. Each contains the same amounts of protein and enzyme **R**. The test tubes are kept at different levels of pH and temperature, as shown in the table. In which test tube will protein digestion be quickest?

	pH	Temperature /°C
A	2	20
B	2	35
C	7	20
D	7	35

- 13 The photomicrograph to the right shows two blood cells.

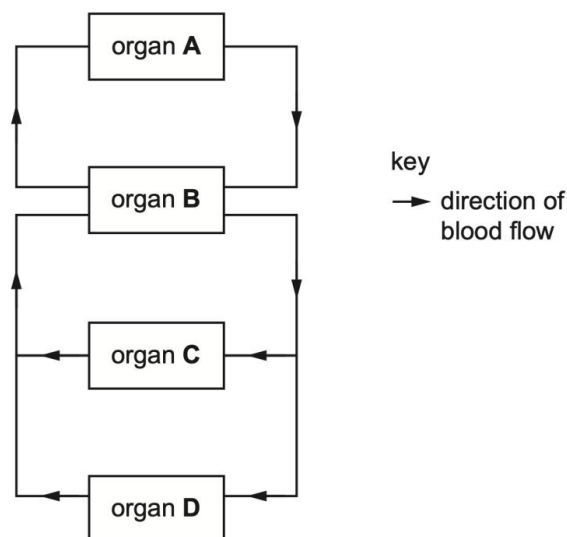
What is the main function of these cells?

- A** to carry out blood clotting
- B** to produce antibodies
- C** to transport carbon dioxide
- D** to transport oxygen



- 14 The diagram shows a simplified circulatory system in humans.

Which organ represents the pump in this circulatory system?



- 15 Which process in living organisms does not use energy released in respiration?

- A** cell division to make new cells for growth
- B** diffusion of oxygen from the alveoli into the blood
- C** maintaining a constant body temperature
- D** muscle contraction to perform physical activities

- 16** Which statement is a correct comparison of aerobic respiration and anaerobic respiration in yeast?
- A** Only aerobic respiration produces carbon dioxide.
 - B** Only anaerobic respiration produces alcohol.
 - C** They both produce water.
 - D** They both use the reactants glucose and oxygen.
- 17** When setting up the chromatography chamber, why must the solvent level be below the pigment spot on the chromatography paper?
- A** To allow the pigments to move faster up the paper
 - B** To ensure the solvent rises evenly up the paper
 - C** To prevent the solvent from evaporating too quickly
 - D** To stop the pigments from dissolving directly into the solvent
- 18** A student investigated the rate of photosynthesis in four species of plants at 20 °C and at 30 °C. All other conditions were kept constant. The results are shown in the table below.

species	rate of photosynthesis /arbitrary units	
	20 °C	30 °C
W	15.5	12.6
X	19.0	30.2
Y	34.5	36.4
Z	18.8	31.7

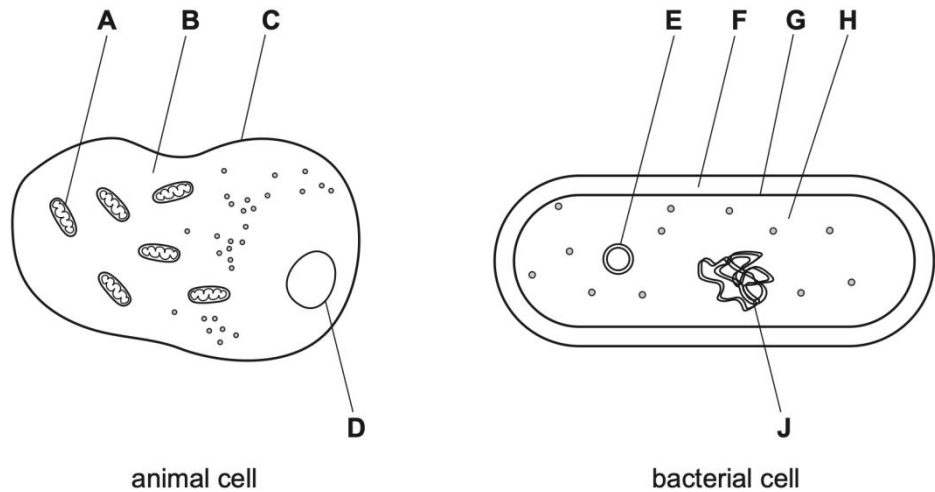
Which statement is a correct conclusion from these data?

- A** Species W photosynthesizes at a lower rate at 20 °C than at 30 °C.
 - B** Species X photosynthesizes at the highest rate at 20 °C and at the lowest rate at 30 °C.
 - C** Species Y photosynthesizes at the highest rate at both 20 °C and at 30 °C.
 - D** Species Z has the greatest change in rate of photosynthesis between 20 °C and 30 °C.
- 19** Which carbohydrate is used to make cell walls?
- A** cellulose **B** glycogen **C** starch **D** sucrose
- 20** Some environmental factors are listed.
- 1 carbon dioxide concentration
 - 2 temperature
 - 3 oxygen concentration
 - 4 wind speed
- Which factors directly affect the rate of photosynthesis?
- A** 1 and 2 **B** 1 and 4 **C** 2 and 3 **D** 2 and 4

Section B Short Answer Questions

- Answer **all** questions.
- Answers must be written **within the answer boxes** provided.

21 The diagram shows an animal cell and a bacterial cell.



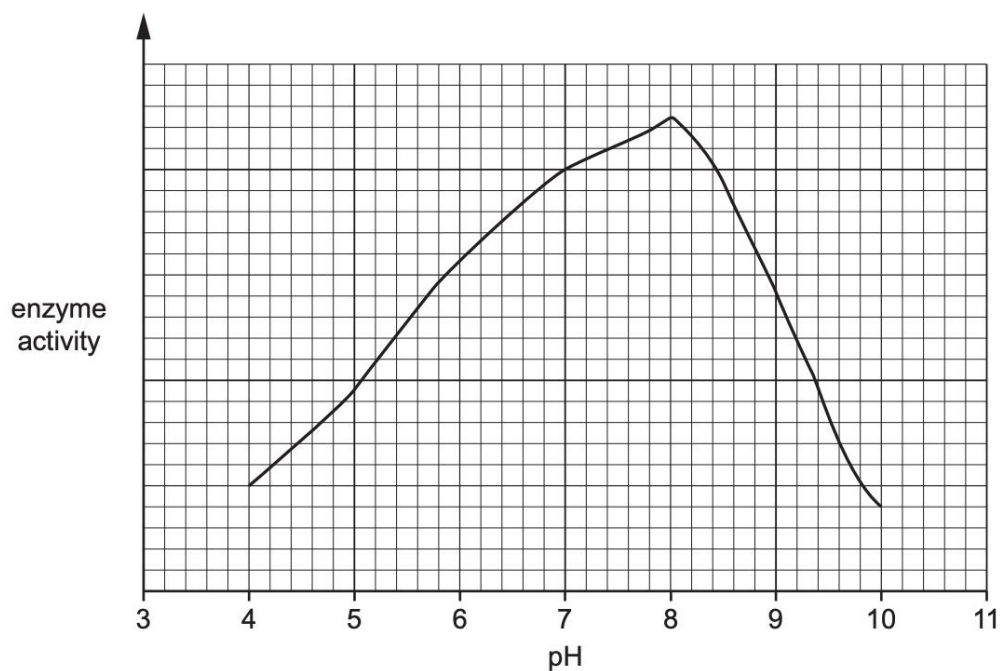
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In the table, **identify** the structures labeled **B** to **J** in the diagram.
Some structures may NOT be present in both cells.

[7]

structure	animal cell	bacterial cell
cell membrane		
cell wall		
cytoplasm		
mitochondria		
nucleus		

- 22 Alcohol dehydrogenase (ADH) is an enzyme that breaks down alcohol in the body. A group of students studied the activity of ADH at different pH values. The graph below shows their results.



- (a) (i) **Identify** the optimum pH for ADH.

[1]

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- (ii) **State** why the activity of ADH is lowest at pH 4 and at pH 10.

[2]

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- (b) Starch is digested by salivary amylase in the mouth, but it is not digested in the stomach.
Suggest one reason for this.

[1]

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.....

- 23 A variegated *Pelargonium* plant was grown outdoors in a plant pot. Figure 1 shows one leaf of the *Pelargonium*. The plant was left in the dark for 24 hours to inhibit photosynthesis. After this time, a sketch was made of the leaf to show the colors (Figure 2), then part of the leaf was covered with black card (Figure 3). Following the exposure of the plant to sunlight for 6 hours, the black card was removed and the leaf was tested for starch (Figure 4).



Figure 1

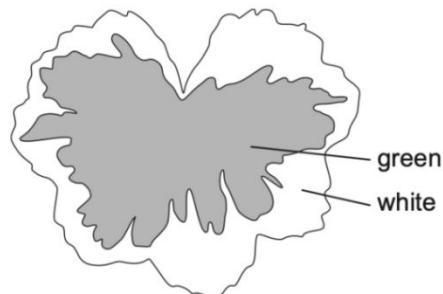


Figure 2

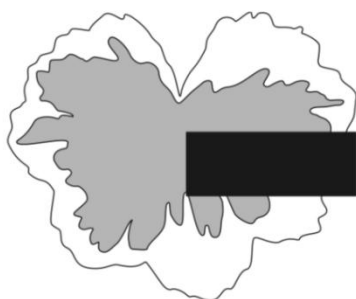


Figure 3

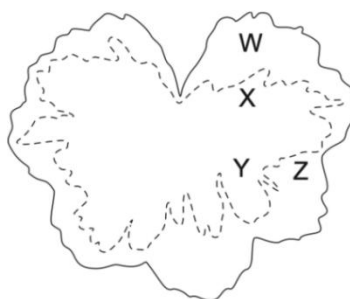


Figure 4

- (a) (i) **Identify** which two areas in Figure 4 show that light is required for photosynthesis. [1]

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- (ii) **Identify** which two areas in Figure 4 show that chlorophyll is required for photosynthesis. [1]

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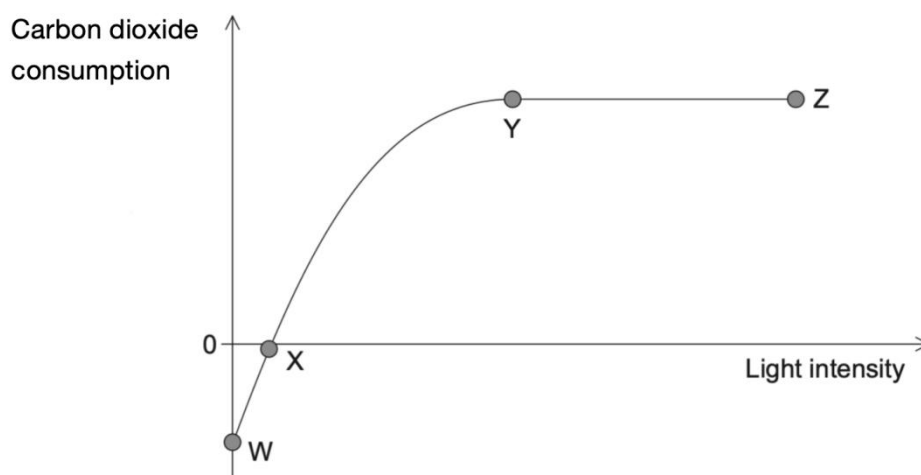
- (iii) **Discuss** briefly whether the detection of starch in this experiment was proof that photosynthesis had occurred in the leaf. [2]

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(This question continues on the following page)

(Question 23 continued)

A group of students investigated the volume of carbon dioxide consumed in a plant at different light intensities. The graph shows their result.



- (b) (i) **Determine** the product of photosynthesis which requires light. [1]

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- (ii) **Explain** carbon dioxide consumption shown by section W–X of the curve. [1]

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- (iii) **Suggest one** limiting factor for photosynthesis at point Z on the graph. [1]

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- (c) **Outline one** reason for inhibiting photosynthesis for 24 hours. [1]

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